

The Model: Feature-Composed Item Factors

SAMPLE (thesis-write Phase C, 2026-08-21) | discussion material, not thesis prose. Follows the three content blocks in 06_fi_protomf.tex §fi-model, written to the model as built (I-host).

Feature-composed I-ProtoMF, fl-ProtoMF, changes the host model in exactly one place. I-ProtoMF [3] represents each item by a free embedding, one learned row per item in a lookup table. fl-ProtoMF replaces this table by a composition. The embedding of item i is the sum of the embeddings of its recorded feature values,

$$\mathbf{q}_i = \sum_{f \in \mathcal{F}_i} \mathbf{e}_f + \mathbf{e}_{\text{ID}(i)}, \quad (1)$$

where $\mathcal{F}_i$ holds the feature values of item i , each $\mathbf{e}_f \in \mathbb{R}^d$ is a learned vector shared by all items that carry the value f , and $\mathbf{e}_{\text{ID}(i)}$ is an optional per-item row whose role is discussed below. The composition follows the established template of feature-aware matrix factorization [1, 2], applied here to the item branch of a prototype model. Everything else remains the host. The item prototypes $\mathbf{p}_1, \dots, \mathbf{p}_K$ stay free learned vectors, the activation of prototype k is still the shifted cosine similarity $t_{i,k}^* = 1 + \cos(\mathbf{q}_i, \mathbf{p}_k)$, the user is still a free vector $\mathbf{u} \in \mathbb{R}^K$ over prototype activations, and the score $S(u, i) = \mathbf{u} \cdot \mathbf{t}_i^*$ is trained with the host’s loss and prototype regularizers (Section 2.3).

Two terms carry this chapter and are fixed here. An attribute is a human-readable catalogue property, the department of a garment or its colour group. A feature is the encoded, model-facing form of such a property, in this model a learned row $\mathbf{e}_f$ entering the sum in Equation 1. Statements about meaning speak of attributes, statements about mechanism speak of features. The model names keep the f of feature rather than attribute because the composition sums whatever encoded rows it is given, an ID handle or an engineered row as readily as an attribute value, and choosing attribute-derived rows is what makes the resulting model interpretable.

The ID row makes the host a special case of the composition. When the feature set $\mathcal{F}_i$ is empty and the ID row is kept, $\mathbf{q}_i = \mathbf{e}_{\text{ID}(i)}$ is again a free per-item vector, and the model reduces to I-ProtoMF exactly. Our implementation reproduces the host bit-identically in this configuration, so every difference observed between fl-ProtoMF and its host is caused by the composed embedding and by nothing else. Switching the two ingredients of the sum yields the three arms evaluated in this chapter. The headline arm, fl-ProtoMF, uses features and the ID row. The noid arm drops the ID row and represents every item from features alone, which isolates what the recorded vocabulary supports by itself. The f0 arm drops the features and keeps the ID row. It is the host reached through the extended code path and serves as the reference point of all comparisons.

The change redraws which components of the model carry meaning. The item embeddings are grounded by construction. Every $\mathbf{q}_i$ is a sum of rows, and each row stands for a catalogue property or for the item’s identity, so the make-up of an item representation can be stated in words. The prototypes remain free learned vectors. They are not grounded themselves, but they now anchor a space populated by attribute rows, and this shared residence is what later allows their meaning to be read from parameters alone (Section 5.4). The user vector remains a purely collaborative object with no vocabulary attached. This division draws a claim boundary the thesis observes throughout. Explanations of this model are stated in attribute vocabulary, because the item side of the score is built from attribute rows. The weights that combine the activations into a score stay learned collaborative quantities, so the model’s reasoning is never claimed to be feature-transparent. On this host the boundary is clean. The activation vector $\mathbf{t}_i^*$ carries the entire dependence of the score on the item, so the grounded vocabulary covers everything the model states about items, while everything it states about the user stays latent by design.

The composition also determines what the model can do with an item it has never seen. Because the score is bilinear, one informative side suffices for a meaningful prediction. A cold item has no trained ID row, but its feature values are known the moment it enters the catalogue, so Equation 1 yields a usable $\mathbf{q}_i$ from the feature rows alone. At cold scoring the untrained ID row is dropped from the sum. Cold users are the mirrored case and stay structurally out of reach for an item-side design, since helping them requires a vocabulary on the user side. Cold-start claims for this model are therefore scoped to cold items met by warm users, a scope that matches the catalogue churn which makes the fashion domain item-cold in the first place. The user side receives its own treatment in Chapter 6.

References

- [1] T. Chen, W. Zhang, Q. Lu, K. Chen, Z. Zheng, and Y. Yu. Svdfeature: A toolkit for feature-based collaborative filtering. *Journal of Machine Learning Research*, 13:3619–3622, 2012.
- [2] M. Kula. Metadata embeddings for user and item cold-start recommendations. In *Proceedings of the 2nd Workshop on New Trends on Content-Based Recommender Systems (CBRecSys) at RecSys*, 2015.
- [3] A. B. Melchiorre, N. Rekabsaz, C. Ganhör, and M. Schedl. Protomf: Prototype-based matrix factorization for effective and explainable recommendations. In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*, pages 246–256, 2022.